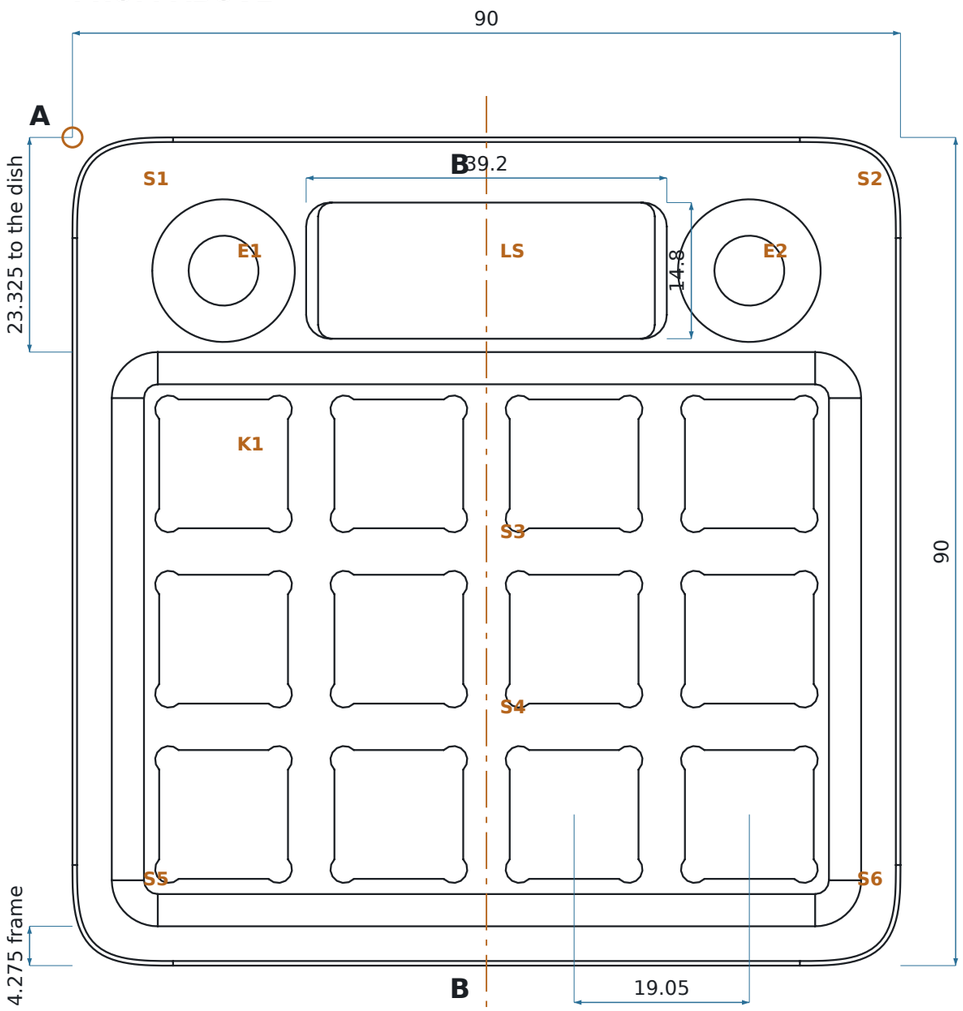
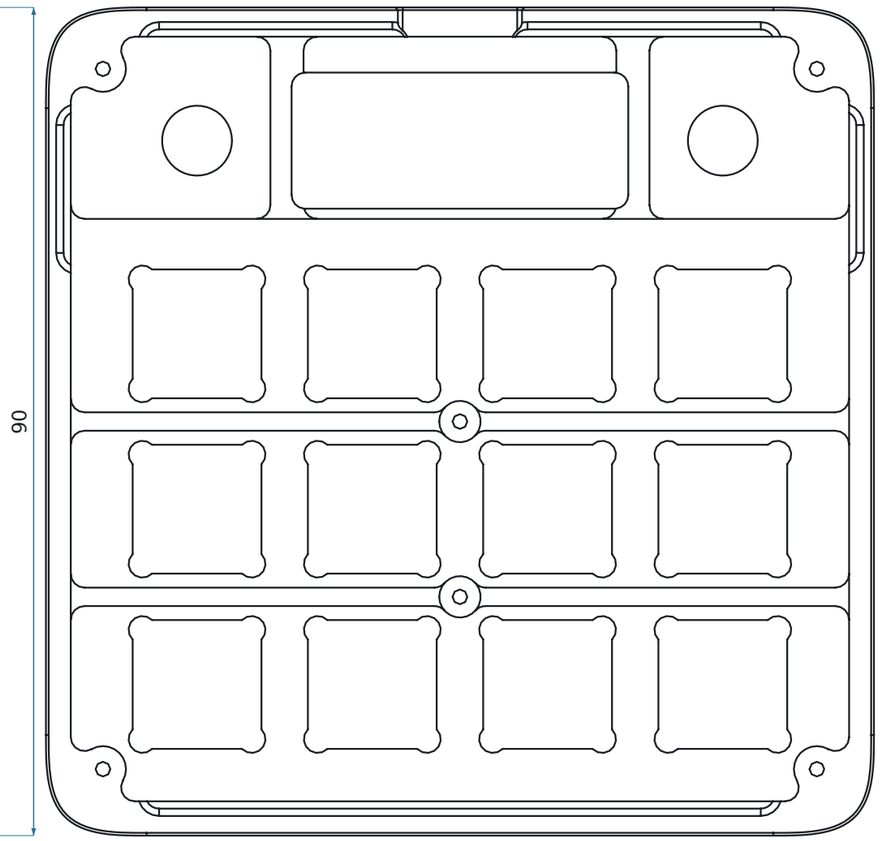
FROM ABOVE



FROM BELOW (mirrored left to right)

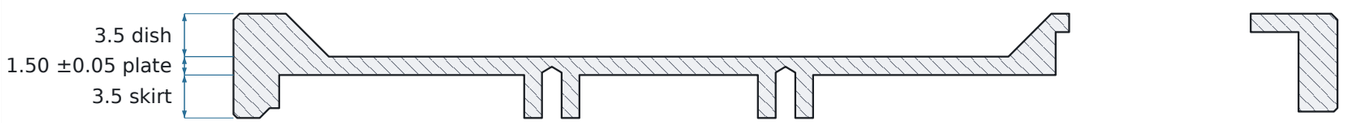


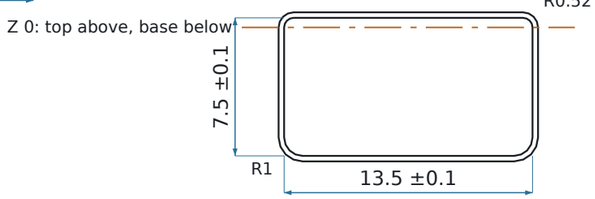
PART	macropad case: top, CNC variant
PART NO.	MP-CASE-TOP-CNC REV A
QUANTITY	1 per macropad (order quantity on the purchase order)
MATERIAL	6061-T6 aluminium
FINISH	Bead blast, then clear anodise (natural silver)
UNITS	mm; plans seen from above and from below, as titled
TOLERANCE	ISO 2768-m unless stated; fits as listed below
STATUS	Not made or measured yet: check the first part

Holes and recesses, from datum A

tag	X	Y
M2x0.4-6H, 2.5 min full thread, from below; position ±0.05		
drill ⌀1.6 x 3.7, do not exceed 3.8 (point to 4.18)		
S1	6.2	6.675
S2	83.8	6.675
S3	45	45
S4	45	64.05
S5	6.2	82.775
S6	83.8	82.775
⌀7.6 through; well ⌀15.5 x 0.7 deep from above		
E1	16.425	14.475
E2	73.575	14.475
14.00 +0.05/0 square through (after anodise), first of 12 on a 19.05 grid		
K1	16.425	35.475
lens recess 39.2 x 14.8, R2.6, 1.25 ±0.05 deep from above		
opening 36.6 x 14.8 through, in its middle		
LS	45	14.475

SECTION B-B

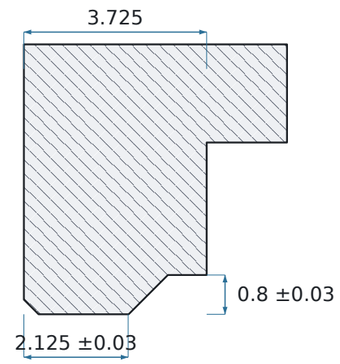




USB OPENING, 3:1

Seen from behind. Rounds along Y: R1 lower (base), R0.52 upper (top). Ball-nose or bull-nose cutter of that radius or smaller, 3D finish; the 3 mm flat end mill cannot cut them. 0.3 x 45 degree lead-in on the outer face.

GAP AND LAP JOINT, 8:1



MATERIAL AND FINISH
6061-T6 aluminium.
Bead blast all over, 120 to 150 grit glass bead, then anodise to MIL-A-8625 Type II, Class 1 (clear, not dyed), 8 to 12 um.
Colour: natural silver, as in the renders. Cosmetic faces: the outside and the top face.
Anodise grows each face by about 0.004 to 0.006 mm (a guess: half the coating). The tolerances below hold after anodise.
Break unmarked edges 0.05. The 0.3 chamfers are cosmetic: measure them after blasting.
Mask the six tapped M2 holes before anodising, or chase them with an M2 6H tap after it.
TOLERANCES (after anodise)
General: ISO 2768-m.
Switch cut-outs 14.00 +0.05/0; plate 1.50 ±0.05.
S1-S6 positions from datum A: ±0.05, on both parts.
Lap grooves: depth 0.8 and land, ±0.03 (detail).
Lens recess depth 1.25 ±0.05; USB opening ±0.1 (detail).
LENS OPENING
Cut it in setup 2, after the lens recess, with the same 3 mm end mill, down to Z 6.35 (2.15 deep). The recess and the opening then share their long walls without a step between setups.

Scale 1.5:1 plans, 2:1 section, 8:1 detail, on A3. Dimensions in mm, computed from the model.
Datum A: back-left corner of the box round the outline, seen from above. X to the right, Y toward the front.
3-axis, two setups: underside up, then top side up.
Every vertical pocket corner is R1.5 or larger; the switch cut-outs have corner reliefs of that radius. The USB opening's rounds are not vertical: see its detail.
Switch cut-outs 14.00 +0.05/0 square, measured after anodise, in a 1.50 ±0.05 mm plate.
Key dish: flat floor on the plate, walls at 45 degrees, 3.5 deep. A 90 degree chamfer mill of 8 mm or more cuts the walls.
Screw holes S1-S6: M2x0.4-6H, 2.5 min full thread, drill ⌀1.6 x 3.7, do not exceed 3.8. For M2 x 8.
Setup 2 (top side up): hold the part on a fixture plate through the six M2 holes, or in soft jaws cut to the outline. The skirts are too thin for vise pressure.
Outline corners: one quintic Bezier each, control points 10.94, 6.56 and 2.19 mm from the box corner on both sides. Take the curve from the STEP file.